

Troubleshooting Network Services Practical

Task 1 — Simulate a DNS Resolution Failure

Scenario

- Use a local hosts-file entry to intentionally map `www.spotify.com` to a non-working private IP address such as `192.0.2.10`.
- The address `192.0.2.10` is reserved for documentation/testing and should not normally provide a real Spotify service.

Windows Procedure

```
Open Notepad as Administrator.  
Open: C:\Windows\System32\drivers\etc\hosts  
Add: 192.0.2.10    www.spotify.com  
Save the hosts file.  
ipconfig /flushdns  
ping www.spotify.com
```

Expected Result

- The computer should resolve `www.spotify.com` to the intentionally configured address `192.0.2.10`.
- The ping may show Request timed out or another unreachable response because the test address does not provide the actual Spotify service.

Troubleshooting / Fix

- Remove the incorrect hosts-file entry.

```
ipconfig /flushdns  
ping www.spotify.com
```
- The normal DNS resolution should then be used again, subject to the network's configured DNS service.

Task 2 — DHCP Incorrect Default Gateway

Simulated Configuration

- Network: 192.168.10.0/24
- Correct gateway: 192.168.10.1
- DHCP server: 192.168.10.2
- Incorrect DHCP default gateway intentionally configured as: 192.168.10.254
- Client receives an IP such as 192.168.10.10/24.

How to Verify the Problem

- On the client, check the assigned gateway with `ipconfig`.
`ipconfig`
- The client should show 192.168.10.254 as its default gateway instead of 192.168.10.1.
- Ping the local gateway and an external destination. Local communication may work while Internet communication fails because the configured gateway is wrong.

How to Fix It

- Open the DHCP server configuration in the simulator.
- Change the default gateway option from 192.168.10.254 to 192.168.10.1.
- Renew the client's DHCP lease.
`ipconfig /release`
`ipconfig /renew`
- Run `ipconfig` again and verify that the default gateway is now 192.168.10.1.
- Test connectivity with ping.

Completed Result

- Correct DHCP gateway: 192.168.10.1
- Client IP example: 192.168.10.10
- The client can communicate beyond the local subnet after receiving the correct gateway.

Task 3 — Devices Can Reach Websites by IP but Not by Name

Diagnosis

- If a device can access a website by its IP address but cannot access it by hostname, basic IP connectivity is working and DNS name resolution is the most likely problem.

Step-by-Step Troubleshooting

```
ipconfig /all
```

- Check which DNS server addresses are configured on the client.

```
nslookup www.google.com
```

- If nslookup fails or reports that the server cannot be reached, test the configured DNS service.

```
ping <DNS-server-IP>
```

- Check whether the configured DNS server itself is reachable.

```
nslookup www.google.com 8.8.8.8
```

- If the public DNS query works but the configured DNS server does not, the local DNS configuration or DNS server is likely the problem.

Solution

- Correct the client DNS server address or repair the local DNS service.
- Renew the DHCP lease if DNS settings are supplied by DHCP.

```
ipconfig /release
```

```
ipconfig /renew
```

```
ipconfig /flushdns
```

- Test the hostname again with nslookup and then open the website in the browser.

Conclusion

- The key evidence is successful IP connectivity combined with failed hostname resolution, which points toward DNS rather than a general Internet connectivity problem.

Task 4 — Client Receives an APIPA Address

Problem

- An APIPA address is in the 169.254.0.0/16 range and is normally self-assigned by a Windows client when it cannot obtain a valid IPv4 address from DHCP.

Possible Causes

- DHCP server is powered off, unreachable, or incorrectly configured.
- DHCP address pool is exhausted.
- Client cannot reach the DHCP server because of a cable, Wi-Fi, VLAN or switch connectivity problem.
- DHCP service is not running.
- Incorrect network configuration or relay configuration prevents DHCP messages from reaching the server.

Step-by-Step Troubleshooting

```
ipconfig /all
```

- Confirm that the client has a 169.254.x.x address and inspect the DHCP Enabled and DNS fields.

```
ipconfig /release
```

```
ipconfig /renew
```

- Attempt to obtain a new DHCP lease.
- Check the Ethernet/Wi-Fi connection and verify link status.
- Verify that the DHCP server is running and that its address pool still has available addresses.
- Check the DHCP scope/network settings, including subnet mask, gateway and address range.
- If the client and DHCP server are on different subnets, verify the required DHCP relay configuration.

```
ipconfig /all
```

- After correction, verify that the client receives a valid address such as 192.168.10.10 rather than 169.254.x.x.

```
ping 192.168.10.1
```

- Test the default gateway and then test an Internet destination if appropriate.

Task 5 — Common DNS and DHCP Issues

DNS Issues

Issue	Symptoms	Possible Cause	Recommended Solution
DNS server unreachable	Websites fail by name; nslookup times out.	Wrong DNS server address, DNS service down, or network path failure.	Check ipconfig /all, ping the DNS server, verify DNS service and correct the configured DNS address.
Incorrect DNS record	One domain resolves to the wrong address.	Wrong A/CNAME record or stale configuration.	Check the authoritative zone/record and correct the record; verify with nslookup or dig.
DNS cache problem	A recently changed hostname continues resolving to an old address.	Stale client or DNS cache.	Flush the client DNS cache and test again; verify upstream/authoritative records.

DHCP Issues

Issue	Symptoms	Possible Cause	Recommended Solution
DHCP server unavailable	Client receives APIPA address or cannot obtain a lease.	DHCP service/server is down or unreachable.	Check server status, network connectivity and DHCP service; renew the client lease.
DHCP pool exhausted	New clients cannot obtain an address.	All addresses in the configured scope are leased.	Expand the scope when appropriate, release unused leases, or create an additional suitable scope.
Incorrect DHCP options	Client has an IP but cannot reach other networks or resolve names.	Wrong gateway or DNS server supplied by DHCP.	Correct DHCP scope options, then renew the client lease and verify with ipconfig /all.

Final Summary

- DNS troubleshooting focuses on name resolution, DNS server reachability, records and caching.
- DHCP troubleshooting focuses on address assignment, scope availability, client-server connectivity and DHCP options such as gateway and DNS.